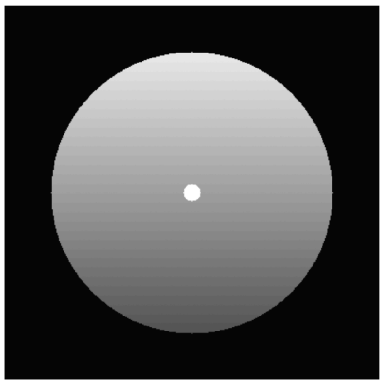
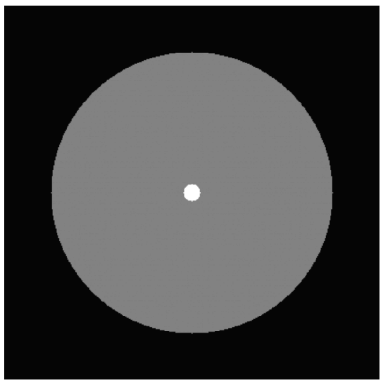
Myopia



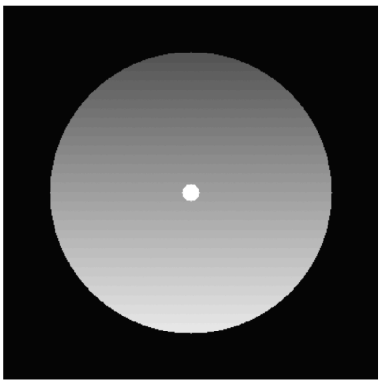
bright region toward top of pupil

Emmetropia



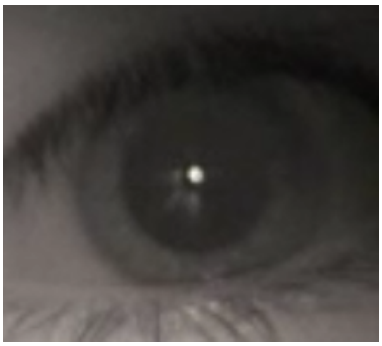
uniform pupil brightness

Hyperopia



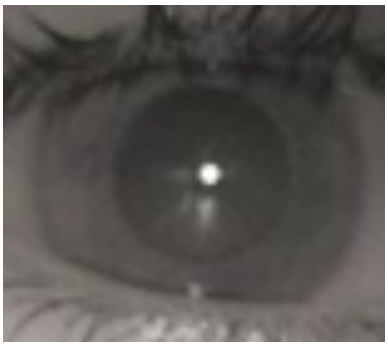
bright region toward bottom of pupil

Myopia



'DIONES.mp4'

Emmetropic

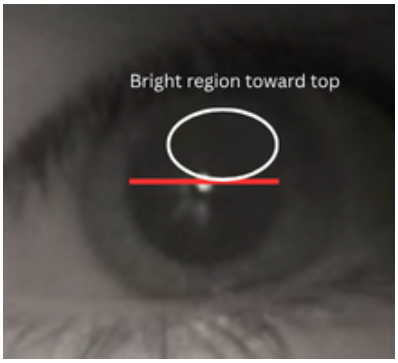


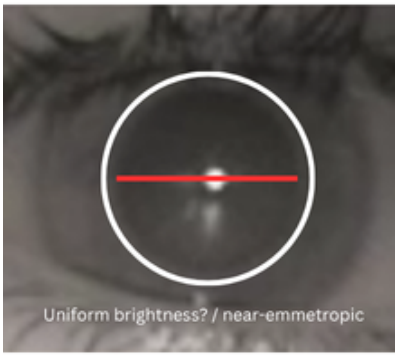
'eye_capture_video10.mp4'

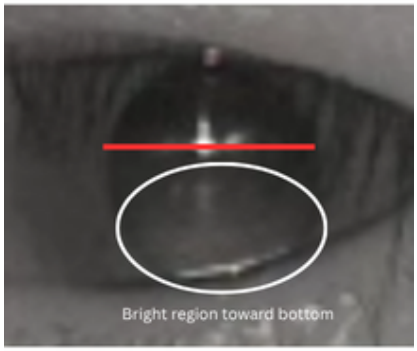
Hyperopia



'high myopia.mp4'







Optical Basis of the Classification Task

The image-based signal that CNNs learn to classify originates from well-defined optical phenomena. In eccentric photorefraction, an infrared light source positioned off-axis relative to a camera aperture causes light reflected from the retina to exit through the pupil as a crescent-shaped brightness pattern; the position and width of this crescent relative to the illumination source indicate whether the eye is myopic, hyperopic, or near-emmetropic (Bharadwaj et al., 2025; Wu et al., 2018). A closely related signal, the red-reflex, is the reflection of light off the retina captured in ordinary flash photography, which similarly shifts in shape and brightness distribution according to the eye's refractive state (Syauqie et al., 2025). In both cases, the diagnostic information is encoded as a spatial pattern within the image rather than as a single scalar value, which is precisely the type of structure convolutional filters are designed to detect.

